

Quiet Asian and Spicy Latino? Designing to Address Racial Microaggression

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ABSTRACT

Racial microaggressions are subtle, often unintentional discriminatory comments that convey negative assumptions about a person's race or ethnicity. It is crucial to address these microaggressions in the design of future technologies, especially speech-based interfaces, as they are known to significantly impact users' acceptance of the technology. However, prior studies in natural language processing (NLP) have highlighted the challenge of detecting microaggressions due to their subtle nature. To tackle this issue, we conducted a survey with 43 individuals to gather insights into common microaggressive comments experienced by people of different races. Based on survey findings, we designed an app called "Inclusify." "Inclusify" encompasses three key features. Firstly, it encourages users to report racist statements and assess their offensiveness, thereby collecting valuable datasets for future speech-based interface design. Secondly, it visualizes common discriminatory expressions through word clouds and showcases trending discussions to raise public awareness. Lastly, the app fosters social belonging by offering a digital space where marginalized individuals can share their stories and concerns. We evaluated the effectiveness of "Inclusify" in increasing awareness through an online survey, comparing users' awareness levels with a pre-post test. Usability was assessed using key task performance metrics and the system usability scale. Based on the results, we discuss design implications and suggest potential directions for future research.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in interaction design**.

KEYWORDS

design for inclusion, racial microaggression, anti-racism, interaction design

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1 INTRODUCTION

Racism is a complex ideology characterized by beliefs in superiority and inferiority based on race [9]. Although the overt form of racism has reportedly decreased over the years, social scientists note that racism in American society has shifted from explicit messages to subtle and implicit expressions [3, 7, 16, 17]. These manifestations are labeled as implicit racism or racial microaggressions, often exhibited by individuals who are not consciously aware of their discriminatory beliefs [16, 17]. For instance, in our survey involving 43 individuals from diverse racial backgrounds, participants reported instances such as labeling blonde white female students as basic, making assumptions about Black individuals enjoying fried chicken, and questioning Hispanic people about their "origin." The survey suggests that racial microaggressions took various forms based on the respondents' race, and all of these instances caused them to feel invalidated and unaccepted. These occurrences are frequently brushed off as innocent mistakes or jokes [16]. The cumulative nature of such racial microaggressions is known to have detrimental effects on individuals, resulting in self-invalidation, social anxiety, and diminished self-esteem [10, 13, 23].

Anti-racism has gained increased attention in the field of human-computer interaction (HCI), particularly in light of the escalating concerns regarding biased data in artificial intelligence (AI) and speech-based interfaces [4, 19, 22]. While extensive research addresses overt forms of racism in natural language processing (NLP), capturing microaggressions in speech-based interfaces remains challenging due to the subtle and covert nature [21]. Implicit racism and microaggressions require developers to grasp the underlying meaning, context, and nuances inherent in language, presenting a challenge in effectively addressing them. To tackle this challenge, we have designed a mobile application called "Inclusify." The Inclusify app has three primary objectives. Firstly, it enables users to share detailed information about their experiences with microaggressions. Users are prompted to rate the level of offensiveness and express their feelings through numeric ratings. This gathered dataset can be utilized for future speech-interface development, promoting more inclusive interactions and communication. Secondly, the Inclusify app generates visualizations of common racist statements heard by different races using word clouds and features relevant stories. As many forms of microaggression may be unintentional, visualizing keywords and reading stories about how specific expressions made others feel can enhance understanding. Lastly, the app provides a space for people to connect through their stories, offering a platform to voice concerns and advocate for change. Our prototype features real stories from 43 diverse voices. Combined, our work aims to contribute to laying the groundwork for the development of more inclusive future technologies.

2 RELATED WORKS

The intersection of race and digital technology has been a fertile research topic in many academic fields, namely Communication, Social Psychology, and Consumer Psychology, along with the Human-Computer Interaction (HCI) community [11]. Despite the increasing interest, there is a notable shortage of research specifically addressing race within the field of HCI [11, 15]. The 2020 survey conducted by Ogbonnaya-Ogburu et al. [11] indicated that, among 30 years of research articles published in CHI, only 120 articles discussed race, amounting to less than 1% of the total. In a more rigorous 2017 meta-review, Schlesinger et al. [15] examined 13,999 proceedings published in CHI, with 140 papers addressing gender, race, and class. In our examination of recent works in the HCI community since 2019, spanning across CHI, CSCW, UIST, IDC, and more, we observed that existing works predominantly concentrate on biased data in technology design [8, 14], disproportionate racial representation in the HCI research community [1, 5], and exploring unique applications of technology by specific racial or ethnic groups [20]. The exploration of how to leverage online communities to foster collective voices on the topic of race, however, is comparatively less extensive [15]. This stands in contrast to the thorough exploration of such an approach in discussions surrounding gender diversity and other marginalized groups [12, 15].

Online platforms have become increasingly crucial as sources for studying racial stereotypes, with social media platforms playing a predominant role in contributing to training datasets in natural language processing (NLP). In a recent meta-review from 2021, Field et al. [6] examined 79 ACL Anthology papers on race and discovered that all of them demonstrated race-related bias in the development of NLP models, highlighting the urgent need for proactive consideration of racism. Field et al. further emphasized that the existing gap is primarily attributed to data sparsity and the prevailing presence of mainstream white American English speakers in NLP research [6]. Within natural language processing, individuals typically have to manually label and classify statements to train the data [22]. The current analysis of these datasets and data training lacks the significant input of racially marginalized individuals, leading to a limited understanding of historically marginalized racial communities [6]. These gaps highlight the necessity of incorporating more diverse voices to share their experiences with racism. Neglecting to tackle racial biases in future technology design might harm users' adoption and trust in the system [14].

Our work aims to address this gap by designing an online platform where individuals can freely discuss their experiences on racism. This not only contributes to generating datasets for future NLP researchers to develop more inclusive interfaces but also provides a space for individuals to voice their concerns, enhancing public awareness.

3 GATHERING REQUIREMENTS

To gain a deeper understanding of the experiences related to racial microaggressions, we conducted in-depth interviews with 4 individuals of various races. Complementing this qualitative approach, we also conducted a survey with 43 respondents. This mixed-methods approach allows us to understand the nuances and varied perspectives comprehensively.

3.1 In-depth Interviews for Problem Identification

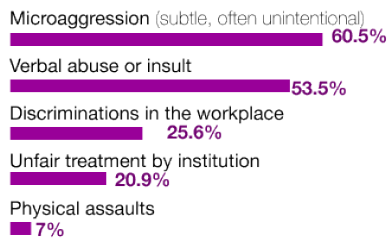
Four students from diverse racial groups, including Black, Asian, and White participated in the interview. Participants were asked to share their experiences of racial discrimination, how such stereotypes made them feel, and how they handled it. A Black female graduate student highlighted the microaggression she experienced:

"Racism towards Black (in higher education) has a different flavor. As a graduate student, people assume I am the type of Black woman they like, so very few people have been explicitly racist towards me. However, people say bigoted things about Black individuals, referring to them as 'general Black.'"

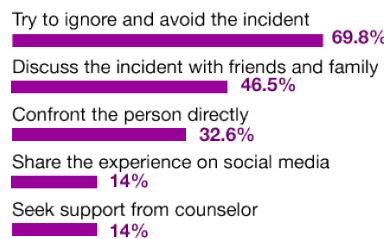
Respondents mentioned that observing implicit racism made them feel excluded and embarrassed. However, when asked how they handle such situations, respondents noted that they typically avoided reporting incidents due to the fear of confrontation and a lack of support, as illustrated in the interview with an Asian student.

"In group projects, other team members didn't give me a chance to present, pointing out my English. I thought about reporting it to the professor, but I was unsure whether he would understand my perspective. Since the professor was also American, I thought he might favor other American students."

What form(s) of racism have you experienced? (multiple choices)



How do you typically respond to incidents of racism? (multiple choices)



What is a racially stereotypical words or sentences have you heard?

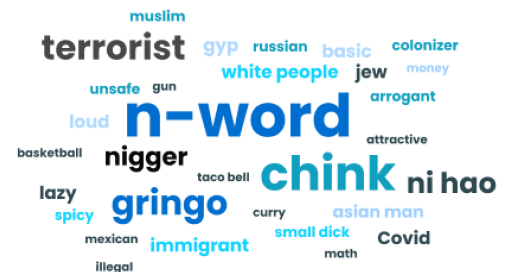


Figure 1: Summary of Survey Results on Respondents' Experience with Racism and Word Cloud of Racist Comments

The interview respondents emphasized the importance of fostering a sense of community. A black female student shared how witnessing the growth of a community of fellow African-American students in academic settings provided her with a valuable sense of support. An Asian student expressed the need for a platform that allows the anonymous reporting of racist incidents. A white female respondent mentioned that she became aware of the strong presence of racial microaggressions after her family adopted a sister with a dark skin color. These findings underscore the essential role of establishing an online community where individuals can share stories and connect to foster better understanding.

3.2 Survey on Experiences with Racial Microaggressions

The survey questionnaire was designed to explore the experience of racial microaggression within a larger population, with the goal of incorporating this data into the design of our app. The survey comprised 25 questions organized into four sections. The first section sought information on the general experience of racism, including inquiries about the frequency of such experiences. The second section featured open-ended questions, asking respondents to share instances of racist expressions and statements they've encountered. In the third section, we presented wireframes of our initial design and asked respondents to provide suggestions through open-ended questions. Lastly, the fourth section of the survey included demographic questions. The survey link was shared through student Discord channels, mailing lists, and online forums. A total of 43 participants with varying demographics and professions responded to the survey. Among the respondents, 37% identified as white, including two individuals who identified as Jewish and one as European. Additionally, 37% of respondents identified as Asian, 13% as Hispanic/Latino, and 9% as Black.

Figure 1 illustrates the summary of our survey findings. 76% of respondents reported that they have experienced racism. The most prevalent form of racism identified in our survey is microaggression, where racially motivated behaviors were perceived as subtle and potentially intentional (60.5%). Verbal abuse ranked second at 53.5%. 62% reported that they have encountered racism at school. However, when asked about how they responded to incidents of racism, almost 70% of the participants indicated that they avoided the confrontation and ignored the incidents. In response to an open-ended question, respondents were asked to list three racist expressions they had heard, resulting in 115 valid responses.

Text analysis was conducted utilizing the Cognos Analytics Toolkit. The term "N-words" and "Nigger" were the most frequently mentioned among Black respondents, followed by other negative connotations, such as being loud, carrying a gun, and living in an unsafe area. Asian respondents highlighted racial slurs such as "Chink" and "ching-chong," with 6 incidents reported. Additionally, random greetings like "ni-hao" were frequently mentioned terms among Asians. Hispanic/Latino respondents indicated that terms such as "Gringo" and "immigrant" were common race-related expressions they had heard. White respondents pointed out that terms like "basic" and "colonizers" were perceived as racially motivated. Jewish respondents reported frequently hearing racially prejudiced expressions. These findings provide valuable insights into the specific forms of racial expressions prevalent among different racial and ethnic groups.

3.3 Evaluating the Early Wireframe

Evaluating wireframes has proven to be an effective approach for engaging users in the early stages of design, encouraging thoughtful and reflective feedback. [18]. Drawing from the interview findings, we initially conceptualized two variations, as shown in Figure 2. The first version showcases different race groups on the main page (e.g., Asian) and allows users to explore common racial slurs or microaggressive expressions encountered by each group (e.g., nerd). In the second version, the emphasis is on "expression" rather than race, presenting keywords first (e.g., nerd) followed by the racial group (e.g., reported by 70% of Asian and 62% of Jews). These initial wireframes were presented to survey respondents, who were tasked with reviewing both versions and providing their suggestions. Among the 43 respondents, 60% indicated a preference for the second version, where the app highlights keywords rather than race. In open-ended questions, respondents pointed out that grouping expressions based on race could lead to division, as illustrated in the following quote: *"I think it's important to highlight the keywords instead of the race. There are overlapping keywords that could harm multiple communities. Division based on race may cause groups to feel divided."*

Respondents also shared their overall thoughts on our approach to addressing racial microaggression. One respondent mentioned that presenting common racist expressions could mitigate microaggressions that occur without malicious intent by increasing public awareness and fostering understanding. We formulated the design

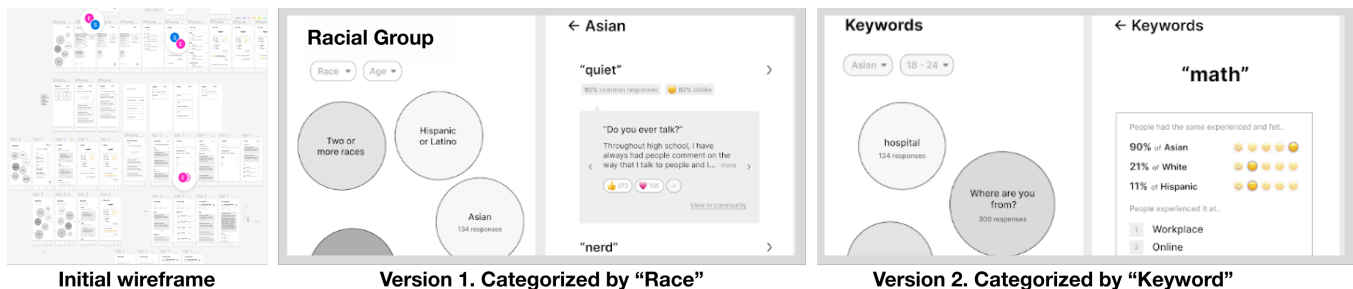


Figure 2: Example images from the early wireframe: We presented two versions to respondents to gather feedback.

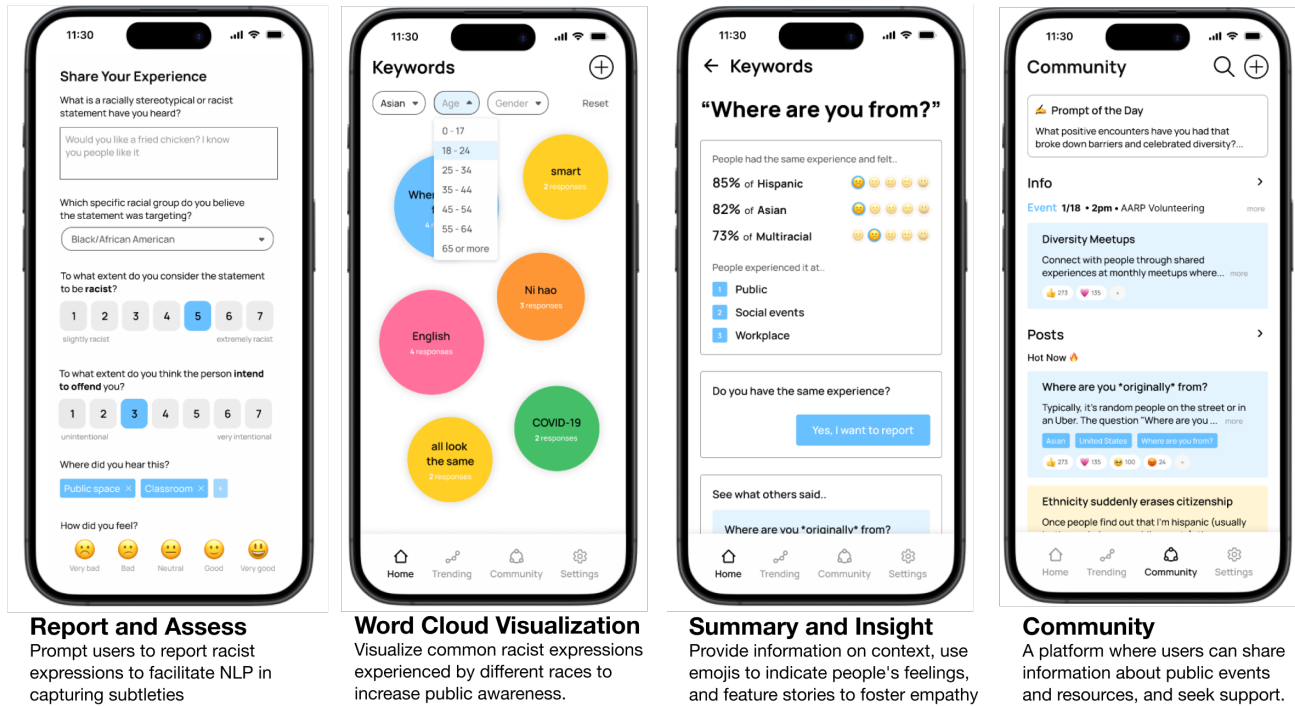


Figure 3: Our Inclusify app encourages users to report racist comments, assess the extent to which these comments felt racist, and express their feelings when exposed to such comments. Utilizing this data, the app visualizes prevalent racist comments and features stories to enhance empathy. The primary objective of the app is to raise public awareness. Furthermore, future natural language processing (NLP) can leverage the collected data to construct a more inclusive speech-based interface.

requirements based on these findings, which are detailed in the following sections.

3.4 Design Requirements

Requirement 1. Comprehensive yet anonymous reporting: Our survey and interview results indicated that most individuals ignore racist incidents to avoid confrontation (70%). Respondents further highlighted that microaggressions are context-dependent. This complexity underscores the importance of users providing comprehensive context when anonymously reporting their experiences with microaggressions.

Requirement 2. Highlight the experience rather than focusing solely on race: Our assessment of the wireframe indicates that prioritizing experiences over racial groups would enhance inclusivity. Featuring stories helps boost empathy by allowing users to listen to the narratives. Based on these findings, we have decided to feature racist comments first, shifting the emphasis from featuring specific racial groups.

Requirement 3. Highlight trending keywords with Likes: The survey and interviews revealed that microaggressions can often be innocent mistakes, as perpetrators might not be aware that such comments could be offensive to certain individuals. These findings indicate the need to help users see ongoing and latest discussions on microaggression. Therefore, we have decided to add a trending tab where users can see discussions that have received the most likes and increasing comments.

Requirement 4. Provide a community: Our interviews suggested that observing people speak out about the presence of stereotypes could support individuals in racial minorities, indicating a need for community voices.

4 FINAL DESIGN: INCLUSIFY

To address identified design requirements, we held a series of brainstorming sessions. We initiated the process with low-fidelity sketches on whiteboards and paper, progressed to a medium-fidelity mock-up, and crafted interactive high-fidelity prototypes using Figma. The final prototype incorporates four features: 1) it prompts users to report and assess experiences, 2) it displays common microaggressions with word cloud visualization, 3) it displays the trending discussions, and 4) it offers an online community. Each design feature is detailed below.

Feature 1. Report and assess The Inclusify app assists users in reporting racist incidents they have experienced by entering the verbal expressions they heard into a text input box, as illustrated in Figure 3. The app also prompts users to rate how offended they felt and evaluate whether they perceive the comment as racially motivated on a numeric scale of 1 to 7. This approach is crucial in addressing microaggressions, as individuals who do not share a similar background may not grasp the subtle nuances and covert nature of microaggressive comments. Studies suggest that including the voices of marginalized individuals in both data reporting

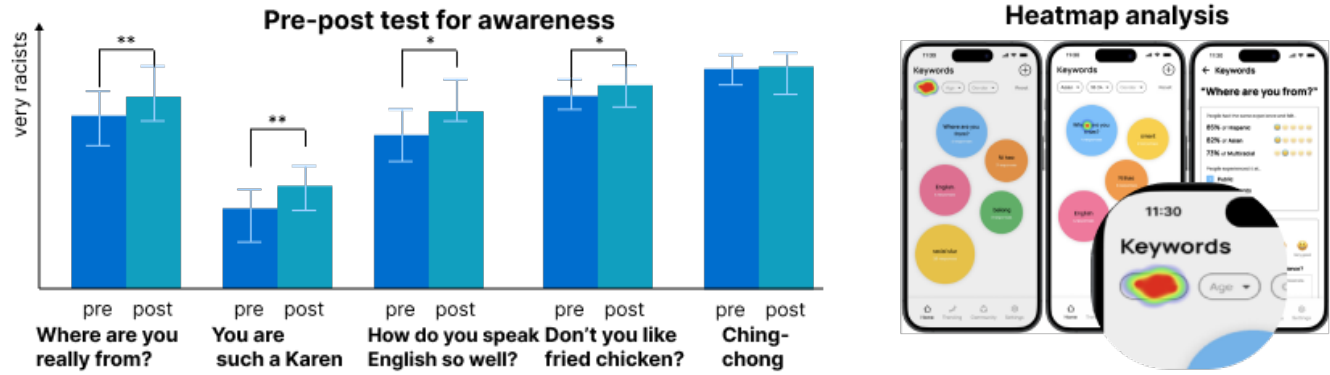


Figure 4: Evaluation) Analysis of Pre-Post Test Results and Heatmaps from Click Tasks (note: ** $p < .01$; * $p < .05$)

and analysis contributes to creating a more inclusive dataset compared to situations where data reporting and analysis are conducted separately, with data analysis primarily conducted by mainstream White American speakers [6]. The app further encourages users to express their feelings using emojis, ranging from feeling good to bad, providing insight into the victim's emotions.

Feature 2. Word cloud visualization The main page of Inclusify visualizes common microaggressions using a word cloud and displays the frequency of such expressions reported to the app. Users can also filter these expressions by race, age, and gender to understand the variations and gain insights into how different racial and ethnic groups experience various stereotypes. When users click on a keyword in the word cloud (e.g., gringo), it directs them to the summary and report page, providing information on how frequently such statements were reported by various racial groups (e.g., X% Hispanic, Y% white). It also features relevant stories under the tab called "see what others say." Our prototype includes actual reports from our survey respondents, with their consent to share.

Feature 3. Trending discussion and keywords The Inclusify app features a "trending" tab that displays real-time updates on trending keywords and discussions. This page highlights discussion forums that have garnered numerous likes and comments. During our initial study, respondents raised concerns about potential issues arising from users with malicious intentions deliberately provoking racist discussions, commonly known as "trolling behaviors" online. To address this, prior works have suggested that incorporating user ratings, such as "thumbs-up" and "heart," could mitigate such ill-intentioned behaviors [11]. Consequently, Inclusify's trending page will factor in the number of hearts and thumbs-up received from the overall user groups in its development.

Feature 4. Community One of the goals of Inclusify is to cultivate a more positive online space where individuals can promote positive behaviors rather than focusing solely on reporting negative experiences. To achieve this, we have included a community tab. At the top of the Community page, a daily "Prompt of the Day" is featured, encouraging users to share positive experiences, such as stories on the growth of African-American female students in engineering communities. Alongside the daily prompt, the Info section displays upcoming events, each with a detailed event post.

This page feature fosters diverse communities, offering users opportunities to establish new connections beyond the app. When users visit the Info page, they can view the three most popular posts and bookmark upcoming events. All these features are carefully designed to ensure easy navigation of the community tab and to find the necessary resources and support, as illustrated in Figure 3.

5 EVALUATION

The evaluation aims to assess the prototype's usability and examine the effectiveness of our approach in enhancing public awareness. To evaluate effectiveness, we conducted a pre-post test where respondents were asked to rate the extent of racism on a scale of 10 (1: not racist, 10: very racist) for five expressions reported as microaggressive in our initial survey ("Where are you really from?"; "You are such a Karen"; "How do you speak English so well?"; "Don't you like fried chicken?"; "Ching-chong"). This approach allows us to measure the participants' perceptions of racism before and after interacting with the Inclusify app, providing insight into the app's potential impact on users' awareness and sensitivity to microaggressive expressions.

To evaluate the usability of the Inclusify prototype, participants were tasked with completing a set of assignments, and we measured both the task completion rate and the time taken for task completion. Specifically, participants were asked to click on a number of keywords, read the corresponding content using Figma and Maze, and then provide feedback. Additionally, we gathered responses using the System Usability Scale (SUS). The survey also included open-ended questions to gather suggestions and feedback on our prototype design.

26 respondents participated in the online survey. 58% of respondents were white, 29% were Asian, followed by 11.8% of Black, and 6% of Hispanic populations. All participants successfully completed the tasks within 10 minutes, indicating ease of navigation [18]. The average SUS score was 73.6, surpassing the recommended criteria [2]. In order to examine the changes in awareness, we conducted independent samples t-tests using R Statistics to compare scores of pre and post-scores for five microaggressive expressions. The results indicated a significant increase in awareness for four out of the five expressions, as illustrated in Figure 4 (e.g., where are you from: $t(25) = -3.43$; $p = 0.002$; $M_{pre} = 7.307$; $M_{post} = 8.12$)

In response to an open-ended question regarding their thoughts on Inclusify, participants noted that the app "*will significantly help people recognize microaggressions*" and "*provides a good platform for individuals within marginalized communities to express frustrations.*" However, several respondents expressed concerns about the potential for individuals with malicious intent to misuse the app.

6 LIMITATION AND FUTURE RESEARCH DIRECTION

The user studies uncovered four primary limitations. Firstly, participants pointed out that the existing categorization system is overly broad, failing to capture the diverse and specific backgrounds of users. One respondent pointed out a need for a more nuanced system that includes subcategories like 'Indian,' 'Vietnamese,' and 'Mongolian' within the overarching 'Asian' community. Overcoming this limitation calls for the development of a categorization system that provides increased granularity and better aligns with the diversity of user backgrounds. Secondly, respondents highlighted a significant concern regarding the potential misuse of the app by individuals with malicious intent, referred to as "bad-faith actors." To address this issue, it is important to implement a robust natural language processing system capable of effectively discerning and filtering out ill-intentioned statements. This measure ensures a secure and constructive environment for users. Thirdly, survey participants noted a potential limitation in the app's ability to reach a broad audience for public awareness. They emphasized that users of Inclusify are likely limited to individuals already concerned about racial microaggression. To overcome this, future projects should explore diverse engagement strategies beyond the app, considering avenues such as public campaigns, alternative platforms, and innovative approaches to broaden the reach and heighten public awareness. Lastly, participants noted a potential limitation in the app's exclusive focus on race. Future research should contemplate expanding the platform's scope to include other marginalized populations in various contexts, such as gender, disabilities, and religion. This expansion aligns with the app's potential for broader application and inclusivity.

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